1 REQUIRED INPUTS

Supply these values first; the equations below then give V.

Loads	D, W_{part}, W_{equip}, L_{storage}, $W_{\text{snow,req}}$
Hazard/site	S_S, S_1, F_a, F_v, T_L
System	R, I_e, C_t, x, C_u
Period	h_n, and T_{calc} if using a calculated period

Order: $W \to S_{MS}, S_{M1} \to S_{DS}, S_{D1} \to T \to C_s \to V$.

2 SYMBOLS

V	base shear	C_s	seismic response coefficient
W	effective seismic weight	D	dead load above base
W_{part}	required partition weight	W_{equip}	operating equipment weight
L_{storage}	storage live load	$W_{\text{snow,req}}$	code snow contribution
S_S	mapped short-period MCE acceleration	S_1	mapped 1 s MCE acceleration
F_a	short-period site coefficient	F_v	1 s site coefficient
S_{MS}	site-adjusted short-period MCE acceleration	S_{M1}	site-adjusted 1 s MCE acceleration
S_{DS}	design short-period acceleration	S_{D1}	design 1 s acceleration
T	period used in direction considered	T_a	approximate fundamental period
T_{calc}	calculated fundamental period	T_L	long-period transition period
R	response modification coefficient	I_e	seismic importance factor
C_t	approximate-period coefficient	x	approximate-period exponent
h_n	structural height above base	C_u	calculated-period upper-limit coefficient
$C_{s,0}$	response coefficient before bounds	$C_{s,\max}$	upper bound on response coefficient
C_{s,S_1}	high-S_1 minimum response coefficient	$C_{s,\min}$	lower bound on response coefficient
g	acceleration of gravity		

3 EFFECTIVE SEISMIC WEIGHT W

$$W = D + W_{\text{part}} + W_{\text{equip}} + 0.25L_{\text{storage}} + W_{\text{snow,req}} \tag{1}$$

4 SEISMIC RESPONSE COEFFICIENT C_s

4.1 DEPENDENCIES

$$\begin{gathered} S_{MS} = F_a S_S, \quad S_{M1} = F_v S_1 \\ S_{DS} = \frac{2}{3} S_{MS}, \quad S_{D1} = \frac{2}{3} S_{M1} \end{gathered} \tag{2}$$

$$T_a = C_t h_n^x \tag{3}$$

$$T = \begin{cases} T_a & \text{hand calc / approximate period} \\ \min(T_{\text{calc}}, C_u T_a) & \text{calculated period available} \end{cases} \tag{4}$$

4.2 COEFFICIENT CHECKS

$$C_{s,0} = \frac{S_{DS}}{R/I_e} \tag{5}$$

$$C_{s,\max} = \begin{cases} \dfrac{S_{D1}}{T(R/I_e)} & T \le T_L \\ \dfrac{S_{D1} T_L}{T^2(R/I_e)} & T > T_L \end{cases} \tag{6}$$

$$C_{s,S_1} = \begin{cases} \dfrac{0.5S_1}{R/I_e} & S_1 \geq 0.6g \\ 0 & S_1 < 0.6g \end{cases} \tag{7}$$

$$C_{s,\min} = \max(0.044 S_{DS} I_e,\ 0.01,\ C_{s,S_1}) \tag{8}$$

$$C_s = \max(C_{s,\min},\ \min(C_{s,0}, C_{s,\max})) \tag{9}$$

CONTROL: use the highest controlling C_s; cap $C_{s,0}$ at $C_{s,\max}$, then compare to $C_{s,\min}$.

5 SEISMIC BASE SHEAR V

ASCE 7 Eq. 12.8-1:

$$V = C_s W \tag{10}$$